

Siddhant Kumar Upmanyu

Senior Systems Architect · Staff-Level Systems Engineer

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EXECUTIVE & STAFF ENGINEERING SCOPE

Hands-on Senior Systems Architect operating with Staff-level engineering scope and sole technical authority over an IoT smart home and commercial platform. Operating under a product manager with zero backend hierarchy; holding single-point custody over cloud infrastructure (Google Cloud Organization Administrator), domain/DNS governance (GoDaddy), on-prem bare metal, and production compute clusters. Specialist in low-level Linux kernel networking, high-throughput telemetry storage, JVM runtime optimization, and autonomous SRE triage. Published distributed systems author and open-source framework maintainer.

TECHNICAL COMPETENCIES

LANGUAGES	Kotlin, Java, Go, Rust, Zig, C, Shell scripting (POSIX/Bash), SQL
KERNEL & NETWORKING	Linux kernel (<code>nftables</code> , <code>iptables</code>), in-kernel NAT64 (<code>Jool</code>), TCP/UDP socket programming, mTLS & custom PKI (CA/CRL), packet reflection, conntrack tuning
DISTRIBUTED & STORAGE	ClickHouse (columnar modeling, Delta+ZSTD codecs), Redis, PostgreSQL, MongoDB, gRPC, Protocol Buffers, REST
CLOUD & SRE	Google Cloud (Org Admin, Cloud Run, Cloud SQL, VPC, GCS, IAM), AWS, Terraform / OpenTofu IaC, Docker, Proxmox VE, GitHub Actions, Linux administration (Debian, Ubuntu)
OBSERVABILITY	Grafana, Loki, Mimir, Alloy, VictoriaMetrics, VictoriaLogs, <code>async-profiler</code> , JMC, VisualVM, Linux performance tools (<code>sar</code> , <code>vmstat</code> , <code>iostat</code> , <code>pidstat</code>)
METHODOLOGY	Extreme Programming (XP), London-Style TDD, Consumer-Driven Contract Testing (Pact), Testcontainers

PROFESSIONAL EXPERIENCE

eGlu Smart Homes (WiZNSystems)

Bengaluru, India · 2022 – Present

Senior Systems Architect (Apr 2026 – Present) | Senior Software Engineer (2024 – 2026) | Software Engineer (2022 – 2024)

Systems Architecture & Kernel Networking

- In-Kernel Dual-Stack NAT64 (`Jool` + `nftables`):** Resolved an edge network partition where cellular IoT hubs operated strictly over IPv6 while backend services were IPv4. Bypassed userspace proxies to preserve CPU credits; deployed the `Jool` kernel module with Explicit Address Mapping Tables (EAMT) alongside `nftables` DNAT to cleanly partition source port ranges on a shared public IPv4.
- Prerouting Packet Reflection & Dynamic Abuse Mitigation:** Offloaded UDP NAT traversal echo responses directly into the Linux kernel using `nftables` packet reflection (`notrack`) at prerouting priority, avoiding userspace context switches. Implemented kernel-level DDoS mitigation via `nftables` dynamic sets (`limit rate over 10/second burst 20 packets` at prerouting priority -301) dropping sweeps before socket allocation.
- Zero-Downtime Kernel Switching:** Designed zero-downtime deployment pipelines using `iptables` DNAT cutovers across both `PREROUTING` and `OUTPUT` chains, explicitly flushing the `conntrack` state table to switch backend traffic without dropping active connections. Hardened recovery scripts after triaging a corrupted `iptables` routing table live during an emergency deployment incident.
- Academic Preprint Publication (Zenodo):** Authored and published *Connection-Agnostic Presence Tracking for Stateless Distributed Backends*: formulated a Redis sorted-set architecture with expiration deadlines and throttled batch writes to track IoT device state with mathematically bounded detection latency.

High-Throughput Telemetry & Storage Systems

- ClickHouse Ingestion Engine (95%+ Compression):** Resolved a filesystem inode exhaustion crisis caused by nested directory log dumps. Designed high-throughput columnar ClickHouse schemas (`FixedString` , `LowCardinality` , Delta + ZSTD codecs, sparse composite primary keys), slashing telemetry footprint by 95%+ (compressing `device_health` telemetry from 17 GB down to 300 MB on disk).
- Decoupled Ingestion Microservice:** Engineered an auxiliary Go microservice communicating via binary gRPC with strict Protocol Buffers contracts to encapsulate ClickHouse, shielding the primary Kotlin/Spring Boot monolith from database driver coupling.
- Binary Payload Decoupling:** Architected and executed a zero-downtime data migration pipeline extracting legacy binary image blobs out of primary database tables, streaming them to secure Google Cloud Storage (GCS) buckets, and swapping schemas to immutable URIs.

Concurrency, Runtime & JVM Performance

- Dynamic IoT-Aware ThreadPool:** Overhauled JVM thread management under bursty IoT packet waves by replacing default thread pools with an adaptive dynamic `ThreadPool` that scales worker threads based on real-time packet velocity and queue depth.
- Deep JVM Profiling & Zero-Allocation Hot Paths:** Instrumented live production systems with `async-profiler` and JMC flame graphs. Mitigated allocation churn with buffer pooling and `ThreadLocal` storage; refactored critical packet decoding loops from `String.format` to Java 17 `HexFormat` and bitwise arithmetic, eliminating hot-path CPU bottlenecks. Pinned heap boundaries (`-Xms = -Xmx` at 4GB) and tuned GC for low pause times.
- Redis Connection Pool Forensics:** Diagnosed and eliminated silent "ghost connections" and thread pool poisoning in Redis Pub/Sub drivers (Lettuce vs. Jedis) that caused cascading application timeouts under transient network partitions.

Platform Engineering, Governance & Cloud FinOps

- **Organizational Cloud Custody & FinOps:** Sole custodian of Google Cloud Organization IAM, VPCs, and GoDaddy DNS zone files. Analyzed multi-year resource utilization to execute a 3-year Google Cloud Committed Use Discount (CUD) on compute instances; currently leading workload evaluation for migrating production workloads from legacy N1 to modern N4 machine types.
- **Production Observability Stack:** Replaced legacy ad-hoc disk logging and custom diagnostic tooling with an enterprise observability stack: Grafana, Loki, and Mimir fed by Alloy with blackbox HTTP/TCP probes, managed under git-tracked configuration (pragmatic GitOps).
- **Zero-Trust Diagnostic Tooling (`hlogger`):** Built a secure client-server log inspection utility in Go using Mutual TLS (mTLS) with a custom PKI hierarchy (Root CA, Intermediate CA, client certificates, CRL), eliminating unauthenticated engineer SSH access to production instances.
- **On-Prem Staging Parity & Bare-Metal Homelab:** Built a dedicated on-prem staging environment replicating production network topology, terminating "testing in production." Configured a Proxmox VE bare-metal cluster to evaluate container orchestration (Kubernetes CNI/CSI, HashiCorp Nomad) for future platform containerization.
- **Declarative Marketing Infrastructure:** Decoupled the public marketing website completely from core backend infrastructure into its own isolated GCP project, codifying 100% of Cloud Run, Cloud SQL (PostgreSQL), networking, and IAM via Terraform / OpenTofu.
- **Platform Automation & Integrations:** Extended scene/rule automation engines for multi-hub environments (fragment handling, cross-hub synchronization); built cloud-to-cloud OAuth integrations for Yale smart locks, Google Home, and Alexa; rewrote hardware device provisioning flows and replacement operations for failed nodes.
- **Technical Hiring:** Formulated backend engineering evaluation rubrics and practical TDD assessments; interviewed and hired software engineering talent for the platform.

OPEN-SOURCE SYSTEMS & RESEARCH

stopgap Kotlin · Maven Central (`2.8.0`)

Microservice framework on Helidon SE (Nima) + Project Loom virtual threads. Custom compile-time DI via KSP (zero runtime reflection). Three-tier testing harness: unit → in-process integration server → Docker E2E via Testcontainers.

assertgo Go · Generics

Type-safe testing assertion library built with modern Go generics. Provides a fluent assertion API, chainable `Not()`, custom matchers, and zero external dependencies.

relay Zig · Systems

Low-level TCP server in pure Zig with a test-driven approach: raw POSIX socket descriptors, manual memory management without libc runtime dependencies, port binding (`SO_REUSEADDR`), and `SIGPIPE` suppression.

hrh Rust · CLI

Helm Release Helper CLI — automated declarative Helm deployments with diff previews and atomic rollback guarantees via `cargo install`.

avoid Shell / Linux · Distro

Minimal, bootable Linux distribution based on Void Linux engineered for server disaster recovery and lightweight headless appliances.

EDUCATION

Bachelor of Computer Applications (BCA)

2019 – 2022